HPS Gaussian-Process Resonance Search
Analysis Note v4.9.12: Expected-Limit Bands and Observed Diagnostics

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Cumulative 100-toy stage. This results-only note uses 100 background-only toys per mass (toy IDs 0–99). Toy IDs 0–49 are the bitwise-preserved initial ensemble and IDs 50–99 are the continuation. The planned 300-toy stage will sharpen the finite-ensemble tails further. All bands and empirical limit-tail diagnostics are pointwise and conditional on frozen GP states.

1 Inputs and construction

The seven scopes are the exact final-sample curves used in the most recent Harvard results section: full 2015, full 2016, the optimized-support 2021 10% sample, their three pairs, and the all-three shared-ϵ^2 combination. The 2021 configuration uses the frozen 36–300 MeV GP support and factor-15 resolution-scaled length ceiling. No 2021 1% or 2016 10% comparison enters these results.

At each mass, a latent background vector is drawn from the conditioned mass-local GP prediction and then Poisson sampled. The same dataset draw is reused by every scope containing that dataset at a fixed mass and toy ID, so the standalone and combined comparisons are paired. GP hyperparameters are not refit inside toys. Each pseudoexperiment is evaluated with the same bounded, piecewise-asymptotic 90% CL_s construction as the observed v4.9.12 curve. For numerical stability, fixed-strength nuisance fits minimize a data-constant-centered Poisson deviance; the fitted likelihood, statistic, and root acceptance gates are unchanged from the parent solver. During the 100-toy continuation, an inherited free-strength optimizer status failed at one deterministic coordinate despite a finite candidate. The contracted solver now uses the same likelihood-equivalent centering as a retry only when a raw free fit is unsuccessful. The released ensemble records 2 centered-profile retry activations (1 free-profile and 1 null-profile); every retry still passes the original likelihood-nesting and root gates, and no toy is dropped or reseeded.

The bands are empirical 2.5%, 16%, 50%, 84%, and 97.5% quantiles. Because different mass points are generated independently, these are pointwise expected-limit bands rather than a scan-wide ensemble. They do not establish unconditional coverage, toy-calibrated CL_s, a look-elsewhere correction, or global significance. Full-2016 and 2016-inclusive curves also retain the previously disclosed cross-process GP-state replay exception.

2 Results

Figure 1 supplies the requested standalone total-window view. It always selects the combination containing every final dataset available at that mass; it does not select among scopes according to the observed or expected limit. Thus the stitching rule is fixed by dataset support rather than by a favorable fluctuation. The displayed 2021 pieces use the optimized-support 10% configuration.

Figure 2 is the direct Brazil-band analogue of the earlier v4.2/v4.5 combination display, now evaluated on the current final inputs. Local motion in the outer envelope should still be read with the finite-ensemble granularity of the current 100-toy stage in mind.

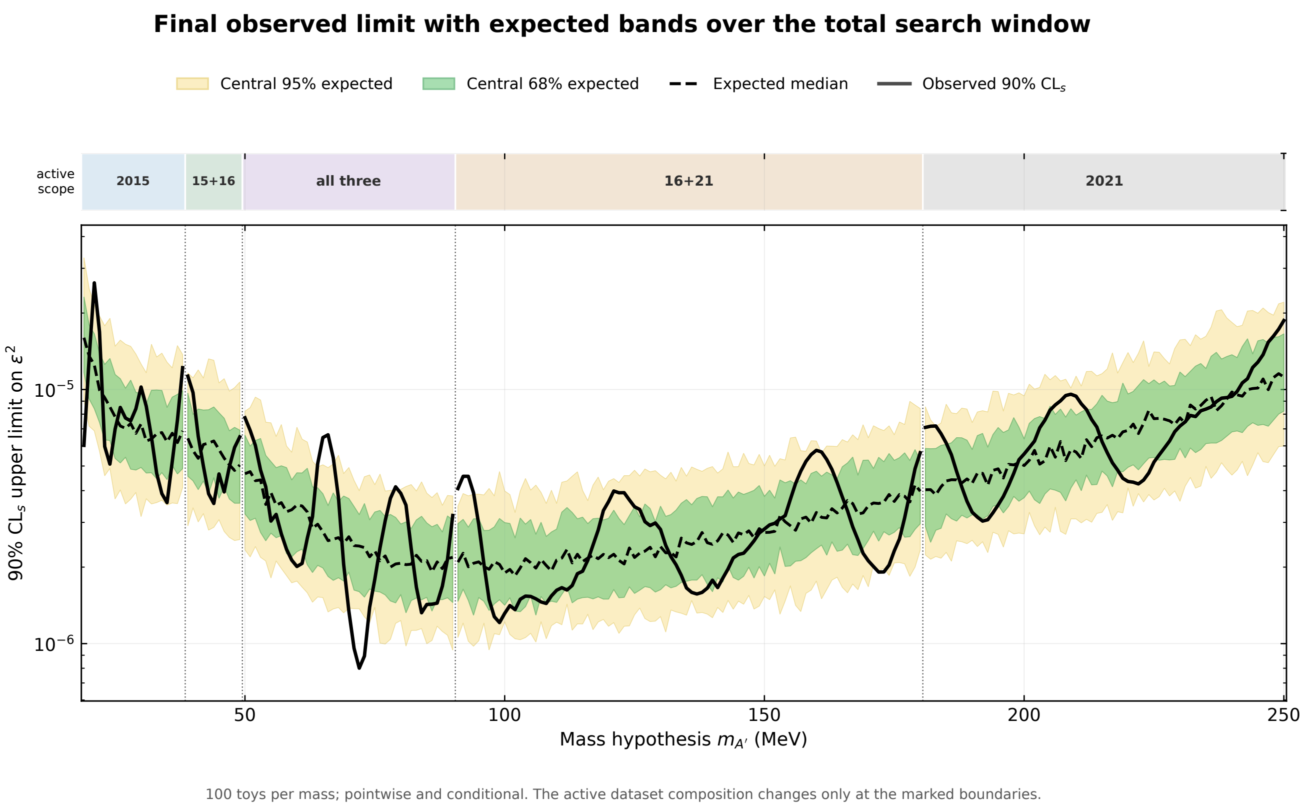

Figure 1: The "final" mass-by-mass maximal-available-dataset observed limit and expected bands over the total search window, 19–250 MeV. The scope strip states which final datasets enter each interval: 2015 at 19–38 MeV, 2015+2016 at 39–49 MeV, all three at 50–90 MeV, 2016+2021 at 91–180 MeV, and 2021 at 181–250 MeV. Scope boundaries are explicit because this stitched display is not one invariant likelihood across the whole range.

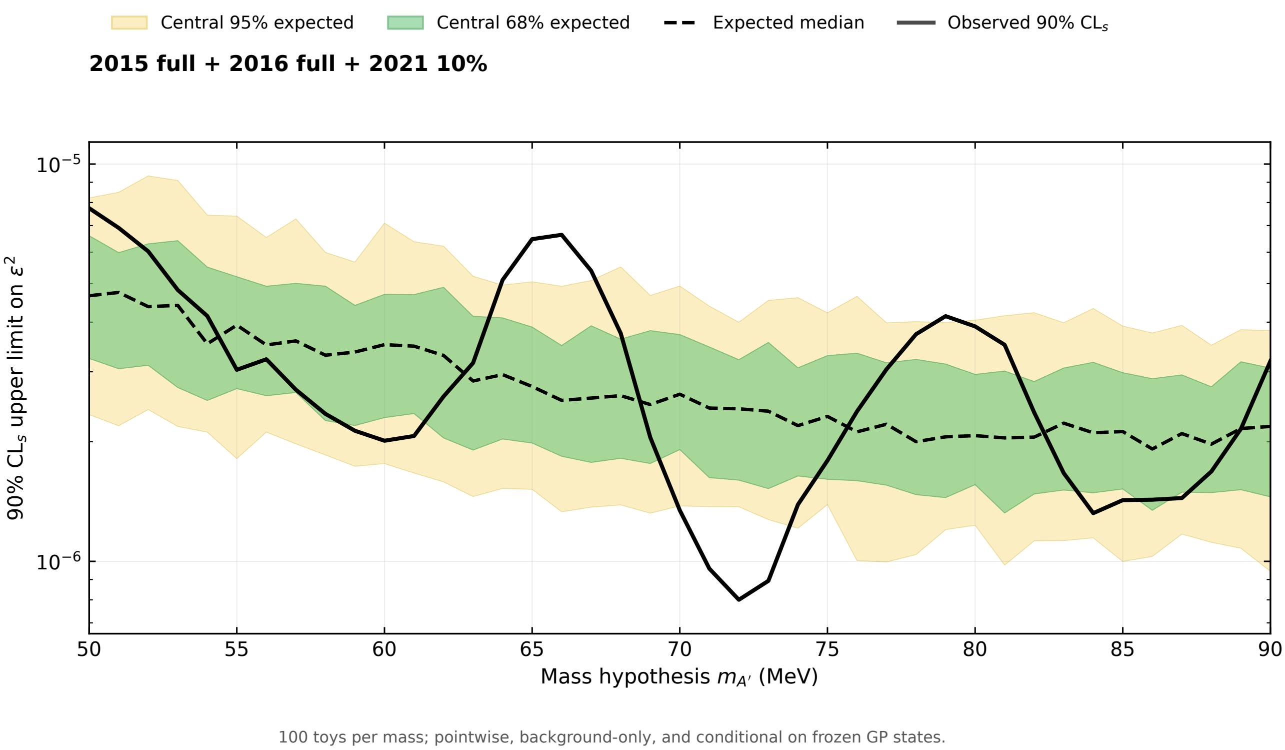

Figure 2: Observed and expected 90% CL_s upper limit for the all-three shared-coupling combination. Yellow and green show the central 95% and 68% pointwise background-only ranges; the dashed curve is the expected median. The legend is outside the data region so it does not obscure any curve.

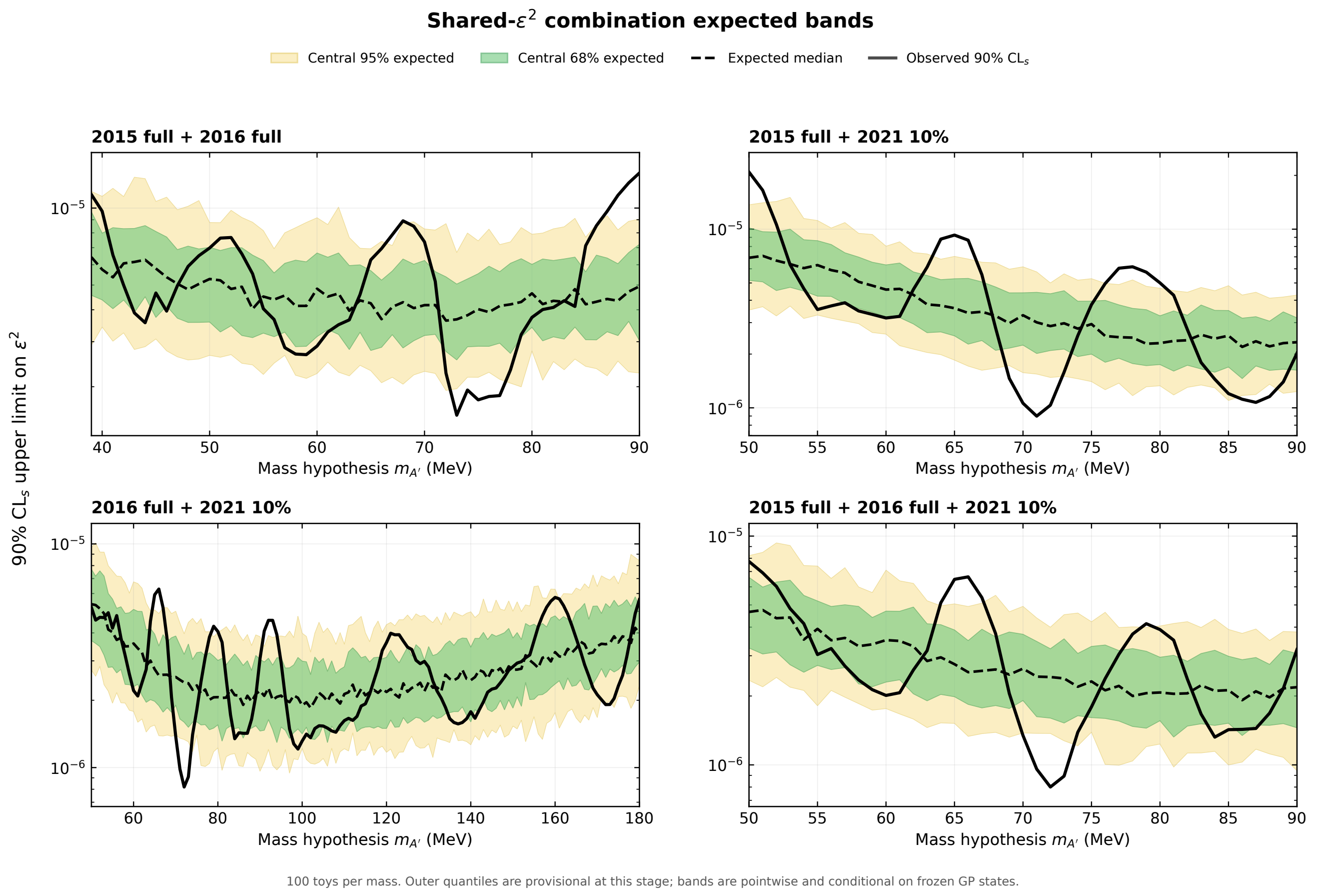

Figure 3: Pairwise and all-three shared-ϵ^2 limits. Each combination has its own axis, preventing bands or observed curves from covering another result. All four panels use paired constituent pseudo-observations.

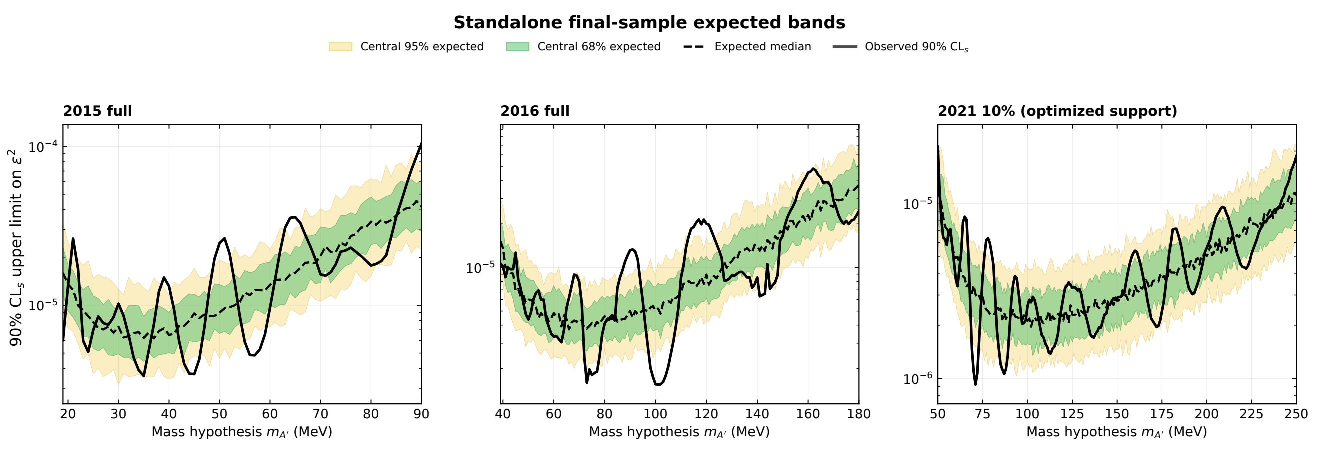

Figure 4: Standalone final-sample limits. The 2021 panel is the optimized native-10% configuration with 36–300 MeV GP support.

2.1 Observed-limit and analytic p-values

Two different diagnostic families are reported and must not be conflated. For an observed upper limit $U_{\mathrm{obs}}(m)$ and toy limits $U_t(m)$ at a fixed mass,

$$p_{\mathrm{strong}}(m) = \frac{1}{N}\sum_t \mathbf{1}[U_t(m) \le U_{\mathrm{obs}}(m)]\,, \tag{1}$$

$$p_{\mathrm{weak}}(m) = \frac{1}{N}\sum_t \mathbf{1}[U_t(m) \ge U_{\mathrm{obs}}(m)]\,, \tag{2}$$

$$p_{\mathrm{two}}(m) = \min[1, 2\min(p_{\mathrm{strong}}, p_{\mathrm{weak}})]\,. \tag{3}$$

A small p_{strong} marks an unusually tight observed limit, while a small p_{weak} marks an unusually loose observed limit. The two-sided number is a symmetric reviewer diagnostic. These are upper-limit ensemble positions, not discovery p-values and not coverage measurements. The one-sided fractions have granularity $1/N = 0.010$, while p_{two} is constructed from them; zero counts are retained as zero in the CSV and shown at half a count only for logarithmic plotting.

The analytic p_0 is separate: it is the frozen observed curve's one-sided, fixed-mass, asymptotic profile-likelihood background-only p-value. It is not derived from these 100 band toys. No look-elsewhere correction is applied, so neither the analytic minima nor the empirical tail fractions support a global-significance claim.

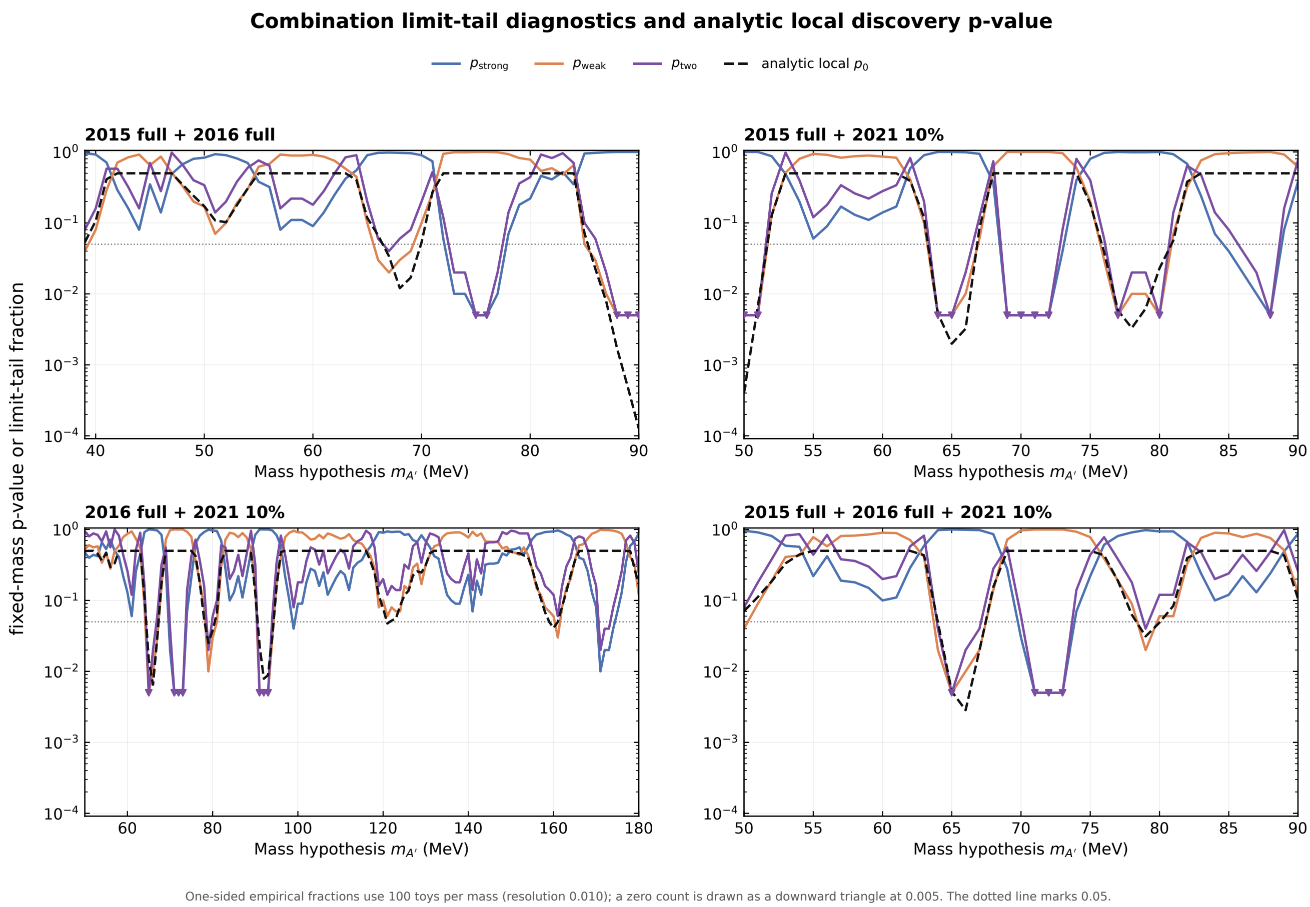

Figure 5: For each shared-ϵ^2 combination, the three empirical observed-limit tail diagnostics and the analytic local discovery p_0. A downward triangle at half the empirical resolution denotes exactly zero toys in that tail; it is not a nonzero p-value estimate. The horizontal dotted line is 0.05 and is included only as a visual reference.

Scope	m [MeV]	p_{strong}	p_{weak}	p_{two}	analytic p_0	Z_{local}
2015 full	51	1	0	0	0.000 846	3.139
2016 full	90	1	0	0	0.000 309	3.424
2021 10%	78	0.99	0.01	0.02	0.002 47	2.810
2015 + 2016	90	1	0	0	0.000 127	3.659
2015 + 2021	50	1	0	0	0.000 403	3.351
2016 + 2021	66	0.99	0.01	0.02	0.0065	2.484
All three	66	0.99	0.01	0.02	0.002 84	2.765

Table 1: Fixed-mass diagnostics at each scope's smallest analytic local p_0. The one-sided empirical fractions have granularity $1/100 = 0.010$ and p_{two} is constructed from them; the selected minimum analytic p_0 is a compact reporting choice, not a trials-corrected global test.

Table 1 gives explicit values at the smallest analytic local p_0 in each scope. Selecting that row makes the compact table scan-aware but does not calibrate the minimum; interpretation remains fixed-mass/local. Complete values for all 680 scope–mass coordinates are in the machine-readable p-value ledger.

2.2 Compact limit summary

Table 2 reports, for each scope, the mass at which the 100-toy expected median is smallest. These minima summarize the plotted curves; they are not optimization inputs and do not select the analysis card.

Scope	m [MeV]	Observed	-2σ	Median	$+2\sigma$
2015 full	37	7.582×10^{-6}	3.558×10^{-6}	6.183×10^{-6}	1.488×10^{-5}
2016 full	66	6.199×10^{-6}	2.151×10^{-6}	3.732×10^{-6}	7.130×10^{-6}
2021 10%	102	2.726×10^{-6}	1.216×10^{-6}	1.977×10^{-6}	4.191×10^{-6}
2015 + 2016	72	2.260×10^{-6}	1.935×10^{-6}	3.628×10^{-6}	8.687×10^{-6}
2015 + 2021	86	1.117×10^{-6}	1.173×10^{-6}	2.193×10^{-6}	4.278×10^{-6}
2016 + 2021	102	1.360×10^{-6}	1.045×10^{-6}	1.850×10^{-6}	3.282×10^{-6}
All three	86	1.429×10^{-6}	1.029×10^{-6}	1.918×10^{-6}	3.755×10^{-6}

Table 2: Observed and conditional expected ϵ^2 limits at each scope's minimum expected-median mass. The outer columns are empirical central-95% endpoints from 100 toys per mass.

Across all 680 scope–mass coordinates, all 68,000 toy limits were finite and closed at CL_s=0.10 within the frozen numerical tolerance. No latent GP draw used the clipping fallback. The exact per-toy limits and pseudo-observation hashes are retained in machine-readable ledgers.

3 Continuation

The ensemble is cumulative. Re-running the driver with `--target-toys 300` appends only IDs 100–299 while retaining IDs 0–99 exactly. Atomic per-mass checkpoints make the continuation resumable, and the contract hash prevents a changed card, GP-state ledger, observed curve, solver, or protocol from being silently mixed into the existing toys.